

Classification and Study of Fradulent and Genuine Reviews Found in e-Commerce Websites

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1 The Problem and Its Relevance to Web Mining

The detection of fraudulent review spamming in the digital landscape, either positively or negatively, is proven to affect the integrity of the perception of a product's quality many times, especially in the e-commerce websites, as reviews in these places are the primary sources that drive a user's purchasing decisions. This problem is heavily deemed to be adversarial, as fraudulent reviews are often times designed specifically to bypass standard fraud-detection systems by mimicking linguistic patterns of genuine reviews.

The project detailed in this proposal is meant to address this fundamental challenge that is encountered many times in the web, as this project will integrate natural language processing with behavioural analysis in order to detect artificially created reviews meant to affect the perception of a product's quality.

2 Project's Goal and the Planned Algorithms

This project's goal is to create a classifier program that can determine if a review about a product in an e-commerce website is genuine or fraudulent, by using three different computational algorithms to compare their performance in detecting a fraudulent review by its signs.

In order to classify the reviews as genuine or fraudulent, I am going to use three algorithms; with Naive-Bayes Classification being the baseline/benchmark algorithm, k-Nearest-Neighbor as the alternative with its lazy learning and distance based classification, and Long-Short Term Memory (LSTM) as the advanced, deep-learning based algorithm that is also used extensively in fraud detection [1].

3 Dataset, Evaulation Procedure and Metrics

In this project, I am going to use the "Fake Reviews Dataset" by Muqadas Ejaz, as it contains 40432 labeled reviews which are balanced in terms of genuine and fake reviews, and contain metadata like category and rating to commit additional studies on, making this dataset ideal for this project.

The evaluation procedure is going to follow a k-fold cross-validation approach, meant for model stability across different product categories. Additionally, a comparative analysis will be performed via the use of a holdout test set to assess the performances of the alternative and advanced algorithms against the baseline algorithm, with the metrics being the industry-standard evaluation metrics of precision, recall and F1-Score to determine the accuracy of the algorithms. The formulas are provided below.

$$\text{Precision} = \frac{TP}{TP+FP}, \text{Recall} = \frac{TP}{TP+FN}$$
$$\text{F1 Score} = \frac{2TP}{2TP+FP+FN}$$

4 Timetable for the Project

This project's deadline is outlined in the proposal guide as "one day before the last lecture of the semester, by 11.59PM (midnight)". Since that date is June 7, the timetable for this project in order to finish it before the timeline is provided below.

Task / Phase	Completion
Data Preprocessing (Cleaning and Vectorization)	April 27
Implementation of Naive-Bayes and k-NN	May 4
Implementation of LSTM Architecture and Training	May 11
Comparative Analysis of Algorithms	May 20
Development of GUI-Based Wrapper	May 29
Final Report and Presentation Prep	June 7

References: [1] MDPI Applied Sciences 2025, 15(21), 11787.